

Wide-Angle Multislice Propagation for Electron Microscopy: Comparison of Fresnel, Angular-Spectrum, and Wave-Propagation Methods

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1. Introduction

Electron microscopy (EM) has recently entered a new era. For a long time, its central role was to produce images of matter, often interpretable only through comparison with image simulations. In this traditional workflow, simulations were used as a forward model: a structural hypothesis was proposed, an image was calculated from this structure, and the agreement with experiment was improved by trial and error. In this sense, image simulation has always been an indispensable tool for extracting reliable structural information from EM data.

This situation is now changing. In modern approaches, especially ptychography and related phase-retrieval methods, the recorded data are no longer simply interpreted as images. Instead, they are used as input for numerical reconstruction algorithms that aim to recover the sample structure, projected potential, or phase shift. In these inverse methods, the electron-matter interaction model is not merely used after the experiment for comparison; it becomes part of the reconstruction itself. Consequently, the accuracy of the forward model is more critical than ever.

A complete description of electron propagation through matter would require solving a many-body quantum-mechanical problem including elastic and inelastic scattering, multiple scattering, and, in principle, backscattering processes. Such a brute-force treatment is not feasible for realistic specimens

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of interest in transmission electron microscopy (TEM). Practical and accurate propagation models are therefore essential. In this work, we focus on multislice approaches, leaving aside Bloch-wave methods, which remain powerful for periodic crystals (author?) [1, 2] but are less general for defects, interfaces, amorphous regions, finite objects, and iterative reconstruction schemes.

The multislice algorithm introduced by Cowley and Moodie (author?) [3, 4] provides such a practical framework. It was originally motivated by a physical-optics picture in which the specimen is represented as a succession of thin phase objects, separated by free-space propagation. In modern terms, the electron wave is alternately multiplied by a transmission function associated with the projected potential of a slice and propagated to the next slice. This simple and efficient construction has become one of the standard tools for simulating electron scattering in TEM.

The conventional implementation of multislice (F-MS) uses a Fresnel, or paraxial, propagator between slices. - - - - - I insert back David's review here - - -

F-MS is usually adequate for conventional high-energy TEM and STEM simulations of thin, low- Z specimens, where other uncertainties such as thermal diffuse scattering, detector response, or microscope aberration characterization often dominate the comparison with experiment. However, for low incident electron energies, thick specimens, or high- Z materials, F-MS can exhibit significant errors because of its paraxial approximation and omission of backscattering (author?) [5, 6, 7, 8]. AS-MS is the minimal FFT-based correction to the paraxial propagation step, but it still treats the specimen as a sequence of thin transmission functions separated by vacuum propagation. More complete wide-angle methods therefore seek to improve the propagation inside the specimen, or to use the full second-order wave equation as a reference.

Further improvements in accuracy, particularly the treatment of large-angle scattering and backward-propagating components, incur significant computational cost. A more accurate electron microscopy multislice algorithm is the Chen–Van Dyck method, also known as Fully Corrected Multislice (FC-MS) (author?) [8, 9]. The complexity of FC-MS arises because the full second-order wave equation contains a square-root operator that mixes a transverse derivative term with the potential and cannot be evaluated as a single Fourier-space propagator.

FC-MS addresses this problem by expanding the square-root operator in a Taylor series and retaining higher-order correction terms that couple

the potential to transverse derivatives. In this way the method accounts more accurately for how the wave evolves within the potential of each slice, rather than treating the slice only as a projected phase grating followed by free-space propagation. This allows improved treatment of larger scattering angles, low-energy electrons, thicker specimens, and high- Z materials and, in its more complete formulation, also gives access to backward-propagating components.

5 years before the FC-MS was introduced in the field of scalar light optics, a solution to the identical mathematical problem of solving a Time Independent Wave Equation to higher accuracy than the F-MS or AS-MS method had already been proposed, known as the Padé Beam-Propagation Method (Padé BPM) (**author?**) [10]. In light optics, the time-independent Helmholtz equation also leads to a square-root propagation operator when it is factorized into forward- and backward-propagating branches. Padé beam-propagation methods approximate this square-root operator using rational functions rather than the Taylor polynomials proposed by Chen and Van Dyck. Because a Padé approximant is a ratio of polynomials, it can represent the square-root dispersion relation more accurately over a wider angular range than a truncated Taylor series of the same order. In practice, this often gives improved stability and accuracy for large-angle propagation, allowing lower-order Padé approximants to achieve accuracy that would otherwise require higher-order Taylor corrections. In either case, approximating the square-root operator remains computationally expensive and incurs significant overhead relative to F-MS or AS-MS.

In electron optics, a still more accurate reference can be obtained by directly integrating the full second-order time-independent wave equation as an ordinary differential equation (ODE). Numerous authors have used such solutions as references for assessing the accuracy of F-MS and the FC-MS method. Wacker and Schröder provide a clear derivation of the ODE and operator-approximation methods and benchmark them against conventional F-MS and direct Runge–Kutta integration of the second-order Schrödinger equation. Their results again showed that the F-MS method can introduce substantial errors at low voltages, large specimen thicknesses, and high scattering angles. Rother and Scheerschmidt approached the problem from a relativistic perspective, using ODE solutions of the Dirac, Klein–Gordon, and relativistically corrected Schrödinger wave equations and comparing them with a paraxial multislice solution. Their work similarly indicates that the dominant high-angle error arises from the parabolic propagation step used in

F-MS. Hillier et al. (author?) [6] show also that the second order derivative term is important to track the positions of Laue Zone rings in low energy CBED patterns of Silicon at 20keV. However, out of all the methods discussed so far, it is only (author?) [8] that attempted to account for high angle effects in a way that can exploit the Fourier transform, with their spherical propagation method. The other series expansion and ODE methods cannot exploit the FFT as directly as split-step multislice, limiting their practical use as a forward model within inverse reconstruction algorithms.

The light-optics community has sought alternatives that retain the computational efficiency of the FFT while avoiding the numerical difficulties of evaluating Padé BPM approximants and still providing a nonparaxial solution to the time-independent wave equation. One such solution to this problem is the Wave Propagation Method (WPM), introduced by Brenner in 1993 (author?) [11]. In the WPM, the potential or refractive index medium is treated as locally homogeneous over small transverse regions, and a corresponding homogeneous one-way propagator is applied for each representative refractive index (see 1 for a visual explanation of the method). The resulting fields are then recombined according to the local refractive-index distribution. This retains the potential-dependent wavelength within the propagation step without explicitly forming the full dense square-root operator. Efficient variants of the WPM, including a recent differentiable formulation, have been developed by (author?) [12, 13]. These methods approximate the nonparaxial wave equation in Fourier space, enabling the use of the FFT while avoiding the need to choose a Taylor-series cutoff, a noted drawback of the FC-MS method (author?) [14]. The WPM is therefore a promising starting point for a practical and more efficient wide-angle multislice method, but to our knowledge this adaptation has not yet been systematically assessed in the electron microscopy context.

This overview motivates the aim of this work: to re-evaluate high-angle propagation methods looking for alternative methods that would be more accurate than the F-MS without being too computer demanding and that thus could be efficiently used in modern inverse EM methods.

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 ? JLR (to improve) using representative thick specimens at high electron energies, and demonstrate an alternative high-angle propagator to the FC-MS method (ignoring backscattering). In particular, we compare AS-MS and WPM against both the traditional F-MS method and a forward-ODE reference solution of the full second-order time-independent wave equation.

We quantify accuracy and computational cost and determine whether either method can provide an efficient correction to paraxial multislice for representative crystalline TEM and STEM simulations.

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This approximation is highly efficient and accurate for small scattering angles, but its limitations become increasingly important when high-angle components contribute significantly to the exit wave, for example in thick specimens, high-Z materials, or strongly scattering objects or low energy EM (author?) [5, 6, 7, 8].

Since modern quantitative and inverse EM methods rely directly on the fidelity of the propagation model, it is timely to revisit this propagation step and assess whether a more accurate yet still efficient alternative can improve the standard Fresnel multislice framework. ----- end of sentences, kept just for thinking

In this work, we compare three multislice (MS) propagators: F-MS, AS-MS, and WP-MS. Section 2 first reviews the wave equation and the main propagation algorithms used to model electron-matter interactions. Section 3 then benchmarks the methods and presents their behaviour in representative CBED simulations. Finally, Section 4 discusses when each propagator method is expected to be adequate.

2. Wave-Equation Models and Propagation Algorithms

This section introduces the scalar wave-equation models and numerical propagation algorithms considered in this work. We first define the Klein-Gordon and relativistically corrected Schrödinger models (Subsection 2.1). We then introduce F-MS and AS-MS, which differ only in their free-space propagation kernel (Subsection 2.2). Finally, we discuss wide-angle and reference propagation methods, including WP-MS and the ODE calculation used for benchmarking (Subsection 2.3).

Throughout this work we use the convention now commonly adopted in the electron microscopy literature, as used for example by Kirkland and Chen-Van Dyck (author?) [9, 15]. e will design the absolute value of the negative electron charge as it is done in kirkland and Reimer; so e is positive. But more importantly, spatial frequencies are measured in cycles per unit length rather than radians per unit length. This convention differs from that used in some older electron microscopy papers, including the original

multislice formulation of Cowley and Moodie (author?) [3], in Wacker and Schröder (author?) [7], and in much of the optics literature (author?) [16]. The vacuum spatial frequency p_0 is therefore

$$k_0 = \frac{1}{\lambda}, \quad (1)$$

where λ is the relativistic electron wavelength. This differs from the optics convention $K_0 = 2\pi/\lambda$, so that $K_0 = 2\pi k_0$.

The consequence of this convention is twofold. First, we define the Fourier transform with the phase factor

$$\mathcal{F}\{f(x, y)\}(q_x, q_y) = \int f(x, y) \exp[-2\pi i(q_x x + q_y y)] dx dy, \quad (2)$$

where (q_x, q_y) are transverse spatial frequencies in cycles per unit length. Second, a plane wave propagating in the direction $\mathbf{k} = (k_x, k_y, k_z)$, with $|\mathbf{k}| = k_0 = 1/\lambda$, is written as

$$\psi(\mathbf{r}) = \exp(2\pi i \mathbf{k} \cdot \mathbf{r}) = \exp[2\pi i (k_x x + k_y y + k_z z)]. \quad (3)$$

2.1. Scalar Relativistic Wave Equations

As demonstrated by Fujiwara (author?) [17] and detailed in Appendix ??, the propagation of high-energy electrons in matter can be described by the time-independent, relativistically corrected Schrödinger equation

$$\left[-\frac{\hbar^2}{2\gamma m_0} \nabla^2 + E_{\text{pot}}(\mathbf{r}) \right] \psi(\mathbf{r}) = \frac{p_0^2}{2\gamma m_0} \psi(\mathbf{r}), \quad (4)$$

where $\psi(\mathbf{r})$ is the spatial part of the electron wave function, $\hbar$ is the reduced Planck constant, m_0 is the electron rest mass, γ is the Lorentz factor, p_0 is the electron momentum in vacuum, and E_{pot} is the potential energy associated with the electrostatic potential V , expressed in volts. With $e > 0$ denoting the elementary charge,

$$E_{\text{pot}}(\mathbf{r}) = -eV(\mathbf{r}). \quad (5)$$

Compared with the non-relativistic Schrödinger equation, the two relevant changes are that the mass in the kinetic operator is the relativistic mass

γm_0 , and that the vacuum carrier wave is fixed by relativistic energy and momentum relations:

$$\begin{aligned} E_0 &= m_0 c^2, & E_{\text{tot}} &= \gamma E_0, \\ E_{\text{kin}} &= E_{\text{tot}} - E_0 = (\gamma - 1)E_0, & \gamma &= \frac{1}{\sqrt{1 - v^2/c^2}}, \end{aligned} \quad (6)$$

$$\begin{aligned} p_0 &= \gamma m_0 v = \frac{1}{c} \sqrt{E_{\text{tot}}^2 - E_0^2} = \frac{h}{\lambda} = \hbar k_0, \\ k_0 &= \lambda^{-1}. \end{aligned} \quad (7)$$

The right-hand side of Eq. (4), $p_0^2/(2\gamma m_0)$, is therefore the eigenvalue of the Schrödinger-like stationary equation; it should not be confused with the total relativistic energy E_{tot} , which appears in the time factor $\exp(-iE_{\text{tot}}t/\hbar)$.

Equation (4) can be rewritten in Helmholtz form,

$$\nabla^2 \psi(\mathbf{r}) + (2\pi k_0)^2 n^2(\mathbf{r}) \psi(\mathbf{r}) = 0. \quad (8)$$

This representation provides a direct link between electron microscopy and the broader optics community, where propagation algorithms are commonly formulated for Helmholtz equations. As explained in Appendix ??, Eq. (8) may be used either with the more general Klein–Gordon refractive index n_{KG} or with its linearised-potential approximation n_{S} . The latter accurately describes the usual electron-microscopy regime, for which $|E_{\text{pot}}(\mathbf{r})| \ll E_{\text{tot}}$.

Introducing the conventional TEM interaction constant (**author?**) [9, 15]

$$\sigma = \frac{2\pi\gamma m_0 e}{h^2 k_0} = \frac{2\pi\gamma m_0 e \lambda}{h^2}, \quad (9)$$

the refractive index used in the corrected Schrödinger regime is

$$n_{\text{S}}^2(\mathbf{r}) = 1 - \frac{2\gamma m_0}{h^2 k_0^2} E_{\text{pot}}(\mathbf{r}) = 1 + \frac{\sigma}{\pi k_0} V(\mathbf{r}). \quad (10)$$

Equivalently, to first order in the potential, $n_{\text{S}}(\mathbf{r}) \simeq 1 + \sigma V(\mathbf{r})/(2\pi k_0)$.

2.2. Multislice Propagators

To obtain conventional multislice from Eq. (8), the wavefunction is written as a rapidly oscillating carrier multiplied by a slowly varying envelope,

$$\psi(x, y, z) = \exp(2\pi i k_0 z) \tilde{\psi}(x, y, z). \quad (11)$$

Neglecting $\partial^2 \tilde{\psi} / \partial z^2$ gives the paraxial equation

$$\frac{\partial \tilde{\psi}}{\partial z} = \frac{i}{4\pi k_0} \nabla_{\perp}^2 \tilde{\psi} + i\pi k_0 [n^2(x, y, z) - 1] \tilde{\psi} = \frac{i}{4\pi k_0} \nabla_{\perp}^2 \tilde{\psi} + i\sigma \tilde{\psi}. \quad (12)$$

For the corrected Schrödinger model, Eq. (10) reduces the interaction term to $i\sigma V \tilde{\psi}$. For slice j , define the projected electrostatic potential and transmission function by

$$V_j(x, y) = \int_{z_j}^{z_j + \Delta z} V(x, y, z) dz, \quad t_j(x, y) = \exp[i\sigma V_j(x, y)]. \quad (13)$$

Fresnel multislice (F-MS) applies this transmission function and then propagates the wave through vacuum with the paraxial Fresnel kernel. Dropping the tilde, one slice is

$$\psi_{j+1}(x, y) \approx \mathcal{F}^{-1} \left[\exp \left(-i\pi \frac{k_{\perp}^2}{k_0} \Delta z \right) \mathcal{F} \{ t_j(x, y) \psi_j(x, y) \} \right]. \quad (14)$$

Each slice requires one FFT pair. The paraxial approximation enters through

$$k_z = \sqrt{k_0^2 - k_{\perp}^2} \simeq k_0 - \frac{k_{\perp}^2}{2k_0}.$$

After removal of the carrier phase, this expansion gives the kernel in Eq. (14). F-MS therefore combines a split-step representation of the specimen with a small-angle approximation to free-space propagation.

Angular-spectrum multislice (AS-MS) keeps the same transmission function and split-step structure, but replaces the Fresnel kernel by the exact free-space dispersion relation (author?) [8, 16]:

$$\psi_{j+1}(x, y) \approx \mathcal{F}^{-1} \left[\exp \left(2\pi i \left[\sqrt{k_0^2 - k_{\perp}^2} - k_0 \right] \Delta z \right) \mathcal{F} \{ t_j(x, y) \psi_j(x, y) \} \right]. \quad (15)$$

Apart from this Fourier-space phase, AS-MS is identical to F-MS and retains the same $\mathcal{O}(P \log P)$ per-slice cost for P transverse pixels. It removes the paraxial free-space approximation, while retaining the thin-slice split-step treatment of the specimen.

2.3. Wide-Angle and Reference Propagation Methods

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1993 (**author?**) [11]. In the WPM, the potential or refractive index medium is treated as locally homogeneous over small transverse regions, and a corresponding homogeneous one-way propagator is applied for each representative refractive index (see 1 for a visual explanation of the method). The resulting fields are then recombined according to the local refractive-index distribution. This retains the potential-dependent wavelength within the propagation step without explicitly forming the full dense square-root operator. Efficient variants of the WPM, including a recent differentiable formulation, have been developed by (**author?**) [12, 13]. These methods approximate the non-paraxial wave equation in Fourier space, enabling the use of the FFT while avoiding the need to choose a Taylor-series cutoff, a noted drawback of the FC-MS method (**author?**) [14]. The WPM is therefore a promising starting point for a practical and more efficient wide-angle multislice method, but to our knowledge this adaptation has not yet been systematically assessed in the electron microscopy context.

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2.3.1. Classical Wave Propagation Method in Scalar Optics

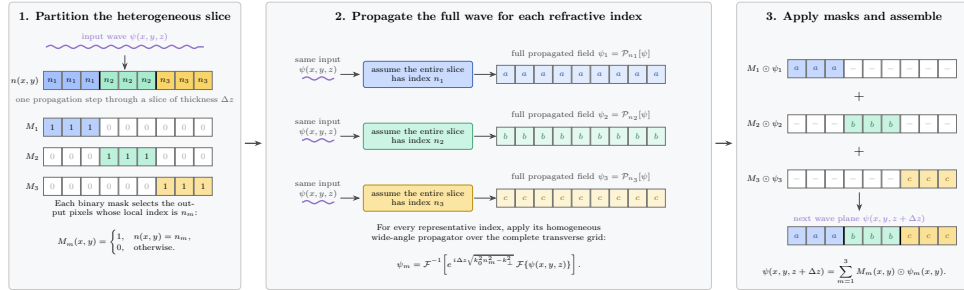


Figure 1: Schematic of the Wave Propagation Method (WPM) as presented by (**author?**) [12]. The input wave is propagated using a bank of homogeneous angular-spectrum kernels, each evaluated at a different representative refractive index n_m . At each output pixel, the propagated field corresponding to the local refractive index is selected via mask and recombined. Our improvement to this algorithm (not described in this figure for simplicity) includes a soft binning step which makes this method suitable for a continuous refractive index distribution - see Figure 2 below.

The Wave Propagation Method (WPM) gathers the split-step approximations in F-MS AS-MS in a one-step propagation, by including the phase shift of the potential in the propagation kernel itself. Indeed, as it is ex-

plained below, starting from the local refractive index of the medium, new local wave vectors $\mathbf{k}(x, y)$ are defined and use in the propagation step.

Returning to the full second-order Helmholtz equation (8), it can be rearranged into an operator equation with a forward and backward branch (author?) [12]. Ignoring the backward branch, the forward branch may be written as

$$\partial_z \psi = 2\pi i \left[\sqrt{k_0^2 n^2(x, y, z) + \frac{1}{4\pi^2} \nabla_\perp^2} - k_0 \right] \psi. \quad (16)$$

The central concern is that the square-root operator inside the wave equation mixes the potential and the transverse derivative, and cannot be evaluated as a single Fourier-space propagator. Padé BPM and FC-MS methods solve for this operator using rational or series approximations to the square-root term, which are expensive and cannot usually directly use the FFT.

Looking again at the AS-MS propagation kernel of equation (15),

$$\left[\sqrt{k_0^2 - k_\perp^2} - k_0 \right]$$

what is primarily missing is a refractive-index term. We can reintroduce this term locally by replacing k_0 with $k_0 n(x, y)$, giving

$$\left[\sqrt{k_0^2 n^2(x, y) - k_\perp^2} - k_0 \right].$$

For a single refractive-index value, this kernel can be evaluated straightforwardly with the angular spectrum method. The WPM simply extends the angular spectrum method by evaluating a bank of homogeneous angular-spectrum propagators for each refractive index in the slice, and then retaining the propagated field corresponding to the local refractive index at each output pixel.

This operation is obviously simplistic, because output pixels can receive a propagated field that travelled from some pixels with the incorrect refractive index. However, the approximation lives in the fact that those different refractive indices are further away from the local refractive-index value of that pixel, and therefore contribute less to the final result. In this manner, the propagation step accounts for the local wavelength inside the specimen without explicitly evaluating the full square-root operator. Schmidt et al. (author?) [12] showed that the leading error from neglecting coupling between

neighbouring refractive-index regions affects amplitude rather than phase; consequently, the WPM phase propagation is correct up to second order.

Figure 1 shows the same local-homogeneous idea in the form used by our electron-microscopy adaptation below. In the classical WPM as implemented by Schmidt, a mask is created to separate out each unique refractive index in a slice, the angular spectrum kernel is calculated for each refractive index, and then the mask is used to choose the correct angular spectrum solution from the bank of all computed propagations.

In Brenner’s original formulation (author?) [11], evaluating a different propagation kernel at every output pixel is prohibitively expensive. For a transverse grid with P pixels, the direct mixed real/Fourier-space calculation scales as $\mathcal{O}(P^2)$. Schmidt et al. (author?) [12] made the method practical for many optical simulations by grouping pixels with the same refractive index. Only one homogeneous angular-spectrum propagation is then required for each distinct index value. If a slice contains M such values, the cost is reduced to approximately

$$\mathcal{O}(MP \log P),$$

instead of a direct all-to-all evaluation.

This saving is large in light-optical simulations where a slice may contain only a small number of unique refractive indices. It is less immediate in electron microscopy, where the electrostatic potential varies continuously and almost every pixel may have a different refractive index. Exact grouping therefore gives little benefit unless the refractive-index field is deliberately discretised. A second issue is that hard grouping introduces sharp masks, which are inconvenient for gradient-based optimisation because the assignment of pixels to refractive-index groups is not smoothly differentiable.

2.3.2. Wave Propagation Multislice

We refer to our adaptation of the WPM suitable for electron microscopy as wave propagation multislice (WP-MS), to distinguish it from the classical WPM used in scalar light optics. To make the computation of WP-MS tractable in the context of electron microscopy, and to avoid the naive $\mathcal{O}(N^{2D})$ calculation for a D -dimensional transverse grid, we adapt partially the WPM algorithm presented in (author?) [13]. Their method makes use of a soft-partition mask between a slice with two unique refractive index values to maintain differentiability of the WPM algorithm. However, they note that they can only use this mask because they have chosen an object with

two refractive index values, and an extension to a more general case would be more challenging.

To adapt this method to a slice with more refractive index values, we first discretise the refractive-index range $[n_{\min}, n_{\max}]$ of the current slice into N_{bins} discrete reference values, denoted as $\{n_{\text{ref}}^{(k)}\}_{k=1}^{N_{\text{bins}}}$.

In a typical multislice potential, many of the refractive-index values will be very close to 1 in the vacuum region, while the refractive-index values inside an atom will concentrate in a narrow range slightly above 1. Since there are likely to be many more refractive-index values near the vacuum region, and the high refractive index values of the atoms are likely to be more sparsely distributed (as the atoms take up comparatively less space in a slice), we employ a quadratic distribution for the reference indices rather than using linearly spaced bins, in order to allocate more bins near $n = 1$. The reference indices are distributed according to a quadratic function determined by the parameter γ , with $\gamma = 2.0$ for quadratic spacing:

$$n_{\text{ref}}^{(k)} = n_{\min} + (n_{\max} - n_{\min}) \left(\frac{k-1}{N_{\text{bins}} - 1} \right)^\gamma, \quad k = 1, \dots, N_{\text{bins}}, \quad (17)$$

which allows for denser binning near the vacuum refractive index when $\gamma > 1$. See 2 for a visual diagram showing graphically the binning step of a slice.

Next, for a propagation step Δz , we compute a bank of propagated fields $u_k(x, y)$. Each field u_k corresponds to the diffraction of the input field $u_{\text{in}}(x, y)$ assuming a homogeneous medium with refractive index $n_{\text{ref}}^{(k)}$. This is performed efficiently using batch FFT operations with the angular spectrum kernel:

$$u_k(x, y) = \mathcal{F}^{-1} \left[\mathcal{F}\{u_{\text{in}}\} \exp \left(2\pi i \Delta z \left[\sqrt{(k_0 n_{\text{ref}}^{(k)})^2 - k_x^2 - k_y^2} - k_0 \right] \right) \right]. \quad (18)$$

To evaluate the field at a specific spatial coordinate (x, y) where the local refractive index is $n(x, y)$, we identify the two bounding reference bins such that $n_L \leq n(x, y) < n_R$, where $n_L, n_R \in \{n_{\text{ref}}^{(k)}\}$. We adopt a differentiable interpolation scheme by defining a local interpolation weight

$$w_{\text{raw}} = \frac{n(x, y) - n_L}{n_R - n_L}, \quad w_{\text{raw}} \in [0, 1] \quad (19)$$

To ensure smooth gradients and minimise discretisation artifacts, we map this linear weight through a smoothstep activation function $S(w)$, akin to the spatial partitioning technique demonstrated by (author?) [13] via a cubic Hermite interpolation function:

$$w = S(w_{\text{raw}}) = 3w_{\text{raw}}^2 - 2w_{\text{raw}}^3 \quad (20)$$

While linear interpolation would satisfy the requirement for differentiability, it could result in discontinuities in the gradient at the bin boundaries in a gradient-based optimisation scheme.

The final field at the next step $u(x, y, z + \Delta z)$ is computed as the weighted superposition of the fields propagated at the bounding indices:

$$u(x, y) = (1 - w) \cdot u_L(x, y) + w \cdot u_R(x, y) \quad (21)$$

Equivalently, the WP-MS update can be written as

$$\psi(x, y, z + \Delta z) \approx \sum_{m=1}^{N_{\text{bins}}} w_m(x, y) \mathcal{F}^{-1} \left[\exp \left(2\pi i \left[\sqrt{k_0^2 \left(n_{\text{ref}}^{(m)} \right)^2 - k_{\perp}^2} - k_0 \right] \Delta z \right) \mathcal{F}\{\psi(x, y, z)\} \right]. \quad (22)$$

This gives a differentiable approximation to the one-way medium propagator in equation (16). When the slice is homogeneous, only one refractive-index bin is needed and WP-MS reduces to the angular spectrum method. For an inhomogeneous specimen, the method is more expensive than AS-MS because many FFT-based propagations are required per slice, but it accounts for the local wavelength inside the specimen rather than propagating every slice with the vacuum wavelength.

This formulation ensures that the propagation model is fully differentiable with respect to the refractive-index map $n(x, y)$, enabling inverse reconstruction of potentials via automatic differentiation. It also means that only a limited number of FFTs, equal to N_{bins} , are required per propagation step, significantly reducing the computational cost compared to a naive WPM implementation. The accuracy of WP-MS depends on the number of bins N_{bins} chosen, and will change on a case-by-case basis depending on the refractive-index distribution of the sample. Figure 2 shows a representative diagnostic of the binning procedure. A practical way to mitigate this potential source of error is to perform a test simulation and double the number of bins until the required tolerance is achieved in the slice error.

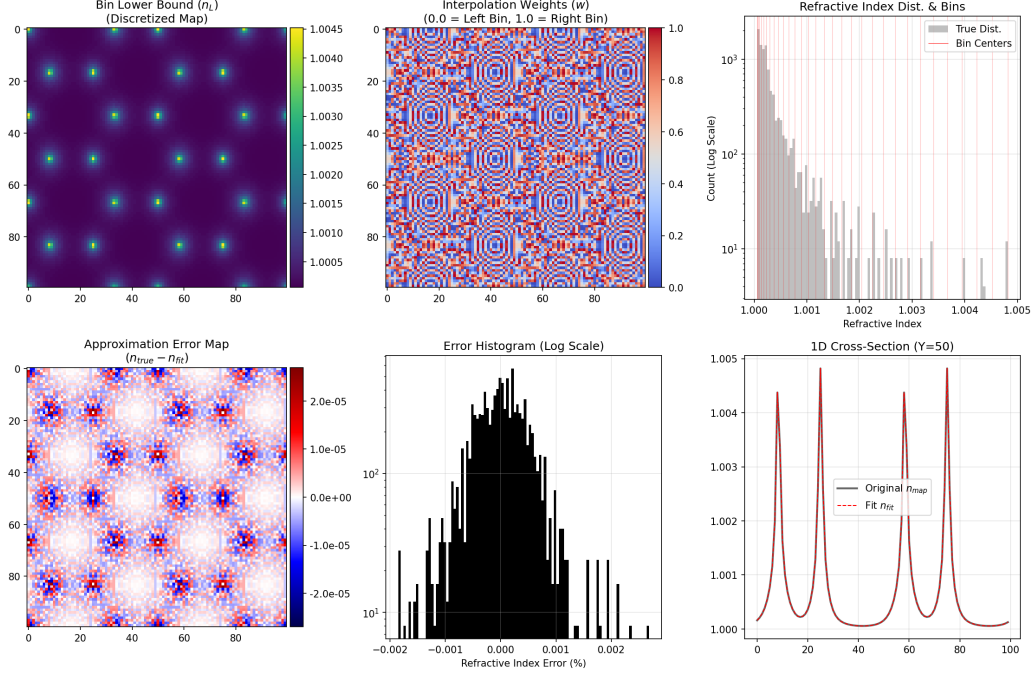


Figure 2: Diagnostics of the differentiable WP-MS binning strategy for a representative MoS₂ slice. Top row: discretized refractive-index map (bin lower bound n_L), smooth interpolation weights w , and histogram of the true refractive-index distribution with superimposed bin centers. Bottom row: spatial approximation error $n_{\text{true}} - n_{\text{fit}}$, error histogram on a log scale, and a 1D cross-section comparing the original and fitted refractive-index profiles.

In general, WP-MS will need a GPU for efficient computation, and we have found that N_{bins} between 16 and 128 are sufficient for the simulations presented in this work. This is a marked improvement over the naive case, as it enables WP-MS to simulate many slices with larger grid sizes (512×512 or 1024×1024 pixels) in a reasonable time frame (seconds to minutes), provided the refractive-index binning strategy is capable of correctly capturing the refractive-index distribution of the sample.

2.3.3. Forward ODE Reference

We use a direct Ordinary Differential Equation (ODE) solution of the second-order Klein–Gordon Helmholtz equation as the reference calculation. The ODE solution is a standard way to solve a wave equation numerically, whereby the transverse coordinates are discretised, the second-order equa-

tion is rewritten as a first-order system, and the resulting large system of ordinary differential equations is integrated along z . Substituting n_{KG} into the Helmholtz equation gives

$$\partial_z^2 \psi = - [\nabla_\perp^2 + (2\pi k_0)^2 n_{\text{KG}}^2(x, y, z)] \psi. \quad (23)$$

Introducing $\phi = \partial_z \psi$ gives the equivalent first-order system

$$\frac{d}{dz} \begin{pmatrix} \psi \\ \phi \end{pmatrix} = \begin{pmatrix} \phi \\ -[\nabla_\perp^2 + (2\pi k_0)^2 n_{\text{KG}}^2(x, y, z)] \psi \end{pmatrix}. \quad (24)$$

We integrate this system with an adaptive high-order Runge–Kutta method. The transverse Laplacian is evaluated spectrally using FFTs, while the potential term is applied by pointwise multiplication in real space.

The calculation is initialised on the forward-propagating vacuum branch. The entrance wave $\psi(z=0)$ is supplemented by the longitudinal derivative

$$\phi(z=0) = \mathcal{F}^{-1} \left[2\pi i \sqrt{k_0^2 - k_\perp^2} \mathcal{F}\{\psi(z=0)\} \right], \quad (25)$$

which selects the positive- k_z branch of the incident field in vacuum. We therefore refer to this as the ODE reference. The method is much more expensive than the split-step propagation methods, although it does include back-scattered effects that occur throughout the interaction process.

The four methods we have presented thus far can be read as successive changes to the same underlying Helmholtz problem, with a chosen refractive index representing either the Klein–Gordon or the relativistically corrected version of the time-independent wave equation, and then an appropriate approximation to the solution of the wave equation. F-MS uses a split-step specimen approximation together with the paraxial free-space propagator. AS-MS keeps the split-step approximation but restores the exact free-space spherical propagator. WP-MS approximates the one-way propagation operator inside the medium by local homogeneous propagation. The ODE solution keeps the second-order Klein–Gordon Helmholtz equation and evolves the transverse Laplacian and specimen potential together, and it is used as the reference calculation in the first numerical experiment. See Table 1 below for a summary timeline of the methods presented in this work.

Method family	Light-optics reference	Electron-optics reference	Description
Fresnel MS	Fresnel; Fleck–Morris–Feit (author?) [18, 19]	Cowley–Moodie; Ishizuka–Uyeda (author?) [3, 4]	Split-step propagation with the paraxial Fresnel kernel.
Angular Spectrum MS	Booker–Clemmow; Korpel et al. (author?) [20, 21]	Ming–Chen (author?) [8]	Same split-step structure, but with the exact free-space propagator.
Padé/Series Expansion (BPMs)	Hadley; Lee–Voges (author?) [10, 22]	van Dyck; Chen–Van Dyck (author?) [9, 23]	Higher-order approximations to the square-root propagation operator.
Wave Propagation Method	Brenner–Singer; Schmidt et al. (author?) [11, 12]	This work	Local homogeneous angular-spectrum kernels applied inside the medium.
ODE	Banerjee–Sharma (author?) [24, 25]	Wacker–Schröder; Rother–Scheerschmidt* (author?) [5, 7]	Direct and reciprocal-space references for non-paraxial propagation.
*Rother–Scheerschmidt used a forward-scattering approximation in Fourier space for a periodic crystal, rather than direct real-space integration of the full second-order ODE.			

Table 1: Historical timeline displaying correspondence between the propagation methods used here and closely related developments in light and electron optics

3. Benchmarks and Results

We present three numerical experiments designed to identify when each propagation method is useful. The first reproduces the Au [100] beam-amplitude benchmark of Rother and Scheerschmidt (**author?**) [5], where F-MS, AS-MS, and WP-MS can be compared with an ODE solution of the full second-order Klein–Gordon equation. After this validation, the ODE reference is not used in the remaining tests because of its high computational cost. The second experiment compares the FFT-based methods for a lower- Z Si [111] CBED calculation at high energy and large thickness. The third experiment considers a thicker Au [100] convergent-probe calculation, using WP-MS as the practical high-angle reference.

3.1. Au [100] ODE Benchmark



Figure 3: The left columns show the real-space exit-wave intensity and phase for the Au [100] verification case. The rightmost column shows the HRTEM image intensity after applying a representative aberration-corrected 300 keV transfer function with residual $C_s = 10 \mu\text{m}$, defocus of +6.0 nm, a 12 mrad objective aperture. Color scales are shared across methods within each column, and image-plane NRMSE values are relative to the Klein–Gordon ODE reference.

To verify our implementation, and to assess which propagation methods retain accuracy at high angles in a thick high- Z specimen, we recreated the simulation presented in figure 3 of (**author?**) [5]. This benchmark propagates a plane wave through an Au [100] crystal at 300 keV and compares against a converged ODE solution of the full second-order Klein–Gordon equation. Periodic lateral boundary conditions are implicit because each method uses Fourier propagation, so the crystal is repeated in the transverse

directions. To test thicker objects, we extended the Rother–Scheerschmidt simulation from 25 unit cells (~ 10 nm) to 100 unit cells (~ 40 nm).

Figure 3 shows the exit-wave intensity, phase, and transfer-function image intensity after propagation through 100 unit cells of Au [100]. The F-MS exit-wave amplitude and phase are qualitatively different from the other three solutions, whereas AS-MS, WP-MS, and the Klein–Gordon ODE reference remain visually similar. After a representative microscope contrast transfer function is applied, however, the differences between methods are much less obvious. This explains why conventional F-MS can still be adequate for qualitative comparison of atomic-column images, even when the unfiltered exit wave contains appreciable high-angle error. To make the comparison more quantitative, Figure 4 plots the normalised Fourier amplitude after each slice for the central $[0, 0]$ component and for the high-angle $[0, 28]$ component at approximately 135 mrad as was demonstrated in the work of (author?) [5].

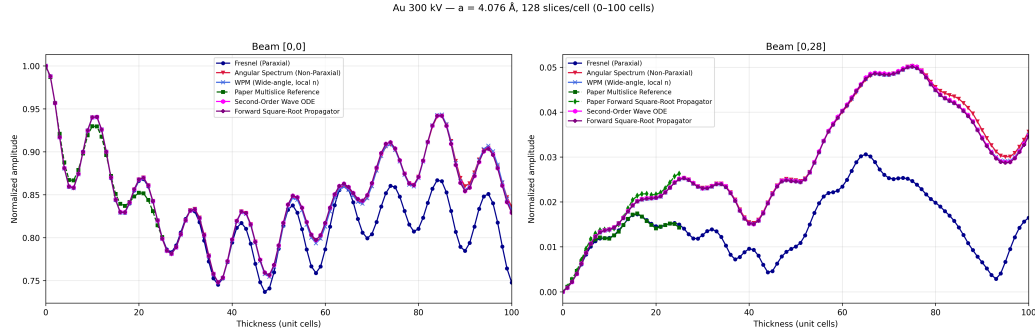


Figure 4: Pixel-amplitude comparison for the Au [100] verification case. The central and high-angle pixels are compared across F-MS, angular spectrum multislice, wave propagation multislice, and the Klein–Gordon ODE reference.

For the central $[0, 0]$ component, all methods agree closely up to approximately 10 nm, or 25 unit cells. Beyond this thickness, the F-MS result begins to deviate from the other methods even on the on-axis pixel. By about 15 nm, high-angle diffracted beams have propagated far enough through the crystal to now interfere back into the central beam, so an error in their accumulated phase can also modify the on-axis amplitude. For the high-angle $[0, 28]$ component at 135 mrad, the F-MS solution diverges from the ODE reference after approximately 10 unit cells, or 4 nm, consistent with the findings of (author?) [5], who attributed the discrepancy to the linearisation

of the square-root operator. AS-MS tracks the ODE reference much more closely than F-MS and remains comparable to WP-MS. The WP-MS solution is slightly closer to the reference, but the difference from AS-MS only becomes clear at the largest thicknesses considered here. These results verify that both the AS-MS method and WP-MS method can adequately track a complete ODE solution to the wave equation, and provide an appreciable improvement over the F-MS method, without the need to solve a higher order Taylor expanded method like the FC-MS solution presented by (author?) [9].

3.2. *Si [111] and Au [100] CBED Simulations*

After verifying AS-MS and WP-MS against the ODE solution, we focus on the three FFT-based multislice methods: F-MS, AS-MS, and WP-MS. The aim is to compare how their predictions differ in CBED patterns for representative low- Z and high- Z specimens. We first simulate Si [111] at 100 keV with an 8 mrad probe and a specimen thickness near 1000 Å, following the spirit of the low-energy CBED comparison of Hillier et al. (author?) [6] but at a higher accelerating voltage. The maximum scattering angle was set to 300 mrad.

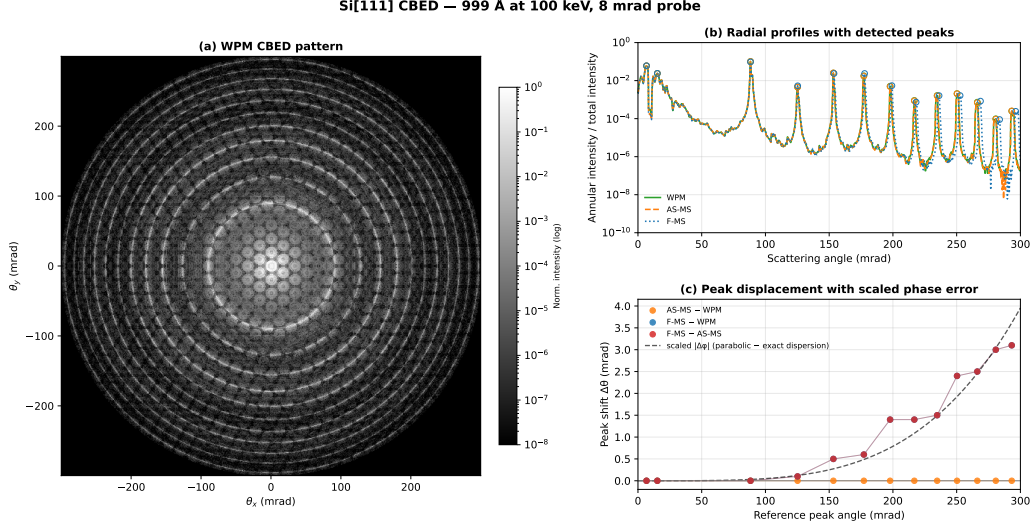


Figure 5: Si [111] CBED comparison at 100 keV, 8 mrad probe, and approximately 1000 Å thickness. (a) WP-MS CBED pattern cropped to 300 mrad. (b) Radially integrated annular intensity for all three methods, with detected diffraction peaks marked. (c) Peak displacement relative to WP-MS, with a scaled parabolic-versus-exact dispersion error curve overlaid. The F-MS—WP-MS shift (blue) follows the dispersion-error curve closely, while the AS-MS—WP-MS shift (orange) remains near zero, consistent with the Au [100] results.

Figure 5 shows the WP-MS CBED pattern, the radial profiles with detected peaks, and the peak displacements. The F-MS peaks shift progressively with scattering angle, but the AS-MS tracks WP-MS closely, with peak displacements near zero across the full 300 mrad range. This suggests that the dominant error in F-MS is the free-space dispersion approximation rather than the split-step treatment of the specimen, even for a lower- Z material at a moderate thickness. (author?) [6] have already identified the need to include the second order derivative of the wave equation to track the Laue zone positions to high angles, and explained using the ewald sphere why there is a difference between traditional F-MS and a second order ODE solution. Our results in part b demonstrate the same effect, but without the need to solve an expensive and slow second order ODE. In part c of 5, we show the x shift of each detected Laue zone peak, overlayed with a rescaled dispersion relation phase error - i.e the difference between a parabola and a sphere. The difference in peaks is proportional to this phase error. The dispersion phase error is computed as $|\Delta\phi| = (2\pi/\lambda) \left| \sqrt{1 - \lambda^2 q^2} - 1 + \frac{1}{2} \lambda^2 q^2 \right|$, where

$q = \sin \theta / \lambda$ is the transverse spatial frequency: the first term is the exact spherical dispersion $\sqrt{1 - x}$ with $x = \lambda^2 q^2$, and the remaining terms are its first-order Taylor expansion $1 - x/2$, i.e. the Fresnel kernel. The curve in panel (c) is this expression fitted to the F-MS – WP-MS peak shifts. The shifted peaks and the fitted curve overlap well, indicating that the error in the output is due to not tracking the spherical propagation between slices in free space. The extra accuracy the WP-MS method provides by tracking the change in wavelength through the potential is not noticeable, and simply swapping the F-MS propagator to the AS-MS propagator already reclaims much of the lost accuracy. It is important to note that in general, thermal diffuse scattering will significantly dampen the presence of peaks beyond 200 mrad, and therefore the application of these methods will be limited for now in traditional TEM, unless perhaps the sample is cooled.

Next, in order to test if a sample with a higher atomic number would highlight more differences between each of the three methods that can use the Fourier Transform, we simulated convergent-beam electron diffraction from an Au [100] supercell at 200 keV using a 5 mrad aberration-free probe. The probe was propagated to a specimen thicknesses of approximately 100 nm again. To sample the high-angle components of the convergent beam accurately, the enlarged 9×9 gold cell was discretized on a 2048×2048 grid. We then propagated the wave using F-MS, AS-MS, and WP-MS, and compared the exit-surface CBED patterns obtained from the Fourier transform of each exit wave.

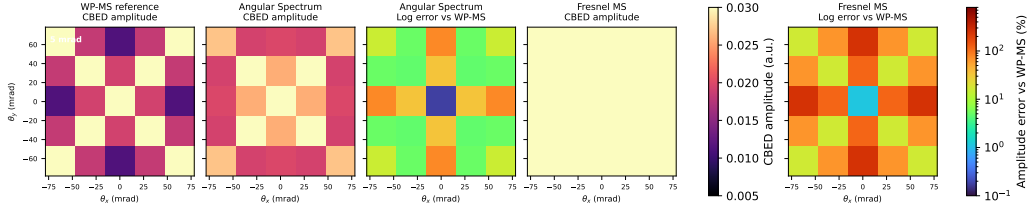


Figure 6: CBED exit-wave amplitude comparison for the Au [100] convergence-angle experiment. The maps show the final diffraction patterns produced by F-MS, AS-MS, and WP-MS, together with log-scale amplitude errors relative to WP-MS.

Figure 6 shows the same pattern in the recorded diffraction plane. The F-MS CBED amplitude differs visibly from WP-MS, especially in the higher-angle features of the CBED pattern. AS-MS remains much closer to WP-MS, and the remaining difference in amplitude between AS-MS and WP-MS is

small compared with the discrepancy introduced by the Fresnel propagator.

Au [100] CBED amplitude relative RMSE vs thickness (reference: WP-MS), 200 keV, 64 slices/cell

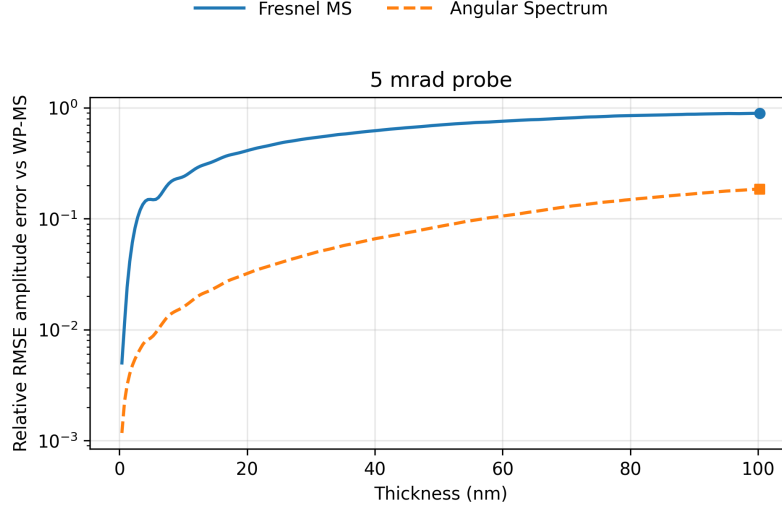


Figure 7: Relative CBED amplitude error as a function of specimen thickness for the Au [100] convergence-angle study. The error is reported against the WP-MS reference at the end of each unit cell.

In Figure 7 we visualise the root mean square (RMSE) error of the CBED amplitude against the WP-MS reference as a function of thickness. After a thickness of approximately 10 nm, there is an order-of-magnitude error difference between the F-MS and AS-MS solutions. The AS-MS method does become less accurate as the thickness increases, but remains much closer to WP-MS than to F-MS throughout the calculation. The residual amplitude difference between AS-MS and WP-MS can be explained as the WP-MS propagation kernel uses the local refractive index, and therefore tracks the potential-dependent wavelength through each slice, decreasing the final error in amplitude. The refractive index of the gold atom near the centre is approximately 1.07, large enough such that over a sample of this thickness, tracking this change can start to materially effect the output CBED image. A similar residual difference in amplitudes between the WP-MS and AS-MS is visible in the exit-wave comparison of Figure 3 from the first numerical experiment, even at 40nm.

Although ptychographic reconstruction for specimens as thick as this is not yet feasible, our results show that the choice of propagator can change

the diffraction-plane amplitudes. Because ptychographic reconstructions are sensitive to systematic amplitude errors, such errors could be absorbed as artifacts or bias the recovered potential. Therefore the choice of propagator may become more consequential as ptychography is pushed toward thicker specimens and higher scattering angles. Given that there is almost no cost associated with the AS-MS method, we at least recommend that it become the default propagator in almost all multislice use cases going forward to minimise another possible source of error in the forward model.

The practical advantage of AS-MS is only immediately useful if it does not require a much finer grid than conventional F-MS. The additional sampling concern comes from the angular-spectrum propagation phase, which is steeper than the Fresnel phase at large transverse wavevector. For one propagation step,

$$\phi_F(k_\perp) = -\pi \frac{\Delta z}{k_0} k_\perp^2, \quad \phi_{\text{ASM}}(k_\perp) = 2\pi \Delta z \left[\sqrt{k_0^2 - k_\perp^2} - k_0 \right].$$

Comparing the phase gradients gives the relative axial sampling penalty

$$\frac{|d\phi_{\text{ASM}}/dk_\perp|}{|d\phi_F/dk_\perp|} = \frac{k_0}{\sqrt{k_0^2 - k_\perp^2}}.$$

This factor is only large when $k_\perp$ approaches k_0 , where the propagating spectrum approaches the evanescent boundary. It can be written directly in terms of scattering angle because $k_\perp/k_0 = \sin \theta$:

$$\frac{k_0}{\sqrt{k_0^2 - k_\perp^2}} = \frac{1}{\sqrt{1 - \sin^2 \theta}} = \frac{1}{\cos \theta}.$$

Under the TEM conditions used here this remains close to unity. For a 200 mrad probe, the largest correction to the Fresnel slice-thickness condition is

$$\frac{1}{\cos(200 \text{ mrad})} \approx 1.02,$$

so the axial sampling bound is tightened by only about 2%. The transverse grid sets a separate limit as it determines the largest scattering angle that can be represented without aliasing. A real-space pixel size Δx gives a

maximum transverse spatial frequency $k_{\perp, \text{Nyq}} = 1/(2\Delta x)$. Since $k_{\perp} = k_0 \sin \theta$ and $k_0 = 1/\lambda$, the corresponding Nyquist angle is

$$\theta_{\text{Nyq}} = \arcsin\left(\frac{\lambda}{2\Delta x}\right) \approx 320 \text{ mrad}.$$

Here we used $\lambda \approx 0.025 \text{ \AA}$ at 200 keV and $\Delta x = 0.04 \text{ \AA}$. This angle is still far from the evanescent boundary at $\theta = 90^\circ$, where the square root in the ASM kernel becomes singular. (Jean Luc needs to add a section here about sampling)

4. Discussion

The discussion part will adress several points. Firstly the interest of the ODE will be discussed. Secondly, the surprising accuracy of WP-MS will be analysed. Thirdly, the comparaison between F-MS and AS-MS will be detailed. The conclusion part will summarize of the important consequences of these discussions and results.

4.1. ODE Discussion

ODE is the best numerical solution to the generalized static Helmoz equation, but in fact there exists many ways to solve an ODE equations. This is a central problem physics and engineering and powerful mathemiatcal and numerical tools have been developp to solve them. In engineering, finite element calculations is certainly the better instrument to solve these kind of ODE. For instance ODE equations are at the core the comsol project and language paradism like python or matlab have develop similar FE solutions. One issue of this approach it is that they are very technical and many physcists and electron microscopists are not specialist of these fields. PROS : black boxes (COMSOL) are available to solve this type of problem (pourquoi ne pas le faire justement avec les licenses du LETI et via Matthieu pour un autre papier ????) CONS : c1) mesh size have to be small to retain all the accuracy c2) consequently if one wants to have a resolution approaching λ , which will be designated as λ -simulations, hudge meshed have to be used and computation time is enormous. c3) it seems difficult to use these approaches even in simple standard λ -simulations of perfect crystals. So using this kind of simulations in phase-retrieval algorithms with a resolution approaching λ , like ptychography is presently unpracticable. This is why more numerical

efficient algorithm like F-MS have to be developed. (6h54 est-ce que je continue d'écrire les idées sur cette partie ODE ? non je crois que j'ai l'essentiel pour l'instant donc je poursuis à la parite suivante NON j'oubliais l'autre idée) c4) Another annoying point is that most of the publications citing ODE resolution of the Helmoz equation does not detail in length the ODE solver that have and it is then difficult to reproduce their result. This is why in this paper we have choosen to give all the details of our ODE solver in the supplementary github site c5) In this paper the ODE solver was used only to validate the 3 other methods on a very simple gold crystal and see the main approximations of the 3 other methods. As ODE results is very close, to other methods even for this high-Z we think it does not need to repeat the ODE for the other systems that are study in this paper.

4.2. *WP-MS approach*

- surprising how it works well in spite of it crude approach ... - explanations could be that (i) it can work well in EM because the index n in matter is rather constant every except at the core the atoms and (ii) it turns out that this approximation is not worse, but even apparently better, than the traditionnal MS. Supposing the wave length in matter is constant is also a very crude and simplistic supposition, but it turns to be working quite well in many situations of EM (see next sub-chapter) - the core of physics is to develop model that mimic the results of experiments. Most of the time, these models are a simplification of what really happen in matter but the interest of the model is not the representation of matter but their habiities to reproduce matter behavior for a given experiment. Models approach the behavior of matter in certain conditions but are not a perfect complete description of matter. A model becomes interesting when it is efficient and simple. - a point interesting to discuss is the change of wavelength that happen in the WP-MS as it also happen in Bloch wave methods. Is that in change in wavelength physical ? Does it really happen in matter ? If yes, how coherent, a key point in ptychography, can be preserved ? This paper will not give an exhausting answer to this question. We will just give the starting following analysis : the change of a wavelength in WP-MS is the just a technical and simple representation of what happens in the experiment and it can be said that in this static representaiton of beam propagation there is indeed a change of wavelength in the piece of sample. But this wave length is not really the wavelength that is linked to coherence : coherence and interference effect does happen in spite of this wave length changes. Cohenrence is a

dynamic effect (?) and what is important in coherence is the pulsation part that varies with time; this part is linked to the energy of the electron and this part is not considered in the present WP-MS static approach.

4.3. *F-MS versus AS-MS*

- x_1 is the key parameter: AS-MS is important only for simulations where x_1 approaches λ , which we have designated as λ -simulations - develop x_1 , g_1 , angles-wave link, evanescent waves ?? - discussion of the singular point at k_0 - but in MS the difference can be more important than in free space propagator (demander à AI pourquoi ? parce qu'il y a en plus de la diffraction à grand angle par les atomes !!! tout simplement)

- up to now, only few papers have performed λ -simulations because it was not needed in TEM to reproduce the experimental images. As resolution in TEM is approaching λ , do we need to need to have more accurate simulation models, especially if someone is not only interested in reproducing visually the experience but want to use this model to reconstruct a 3D-model of the sample. - - in experimental simulations, the wave-propagation model is not the only limiting factors. Energy spread of the incident beam, atom vibrations, sample and electronic drifts, lense and detector imperfections will have their own contribution to the what is called the stobb-factors in EM. - the purpose of these papers was not analysed all these effects but give (i) a short picture of the different efficient numerical approach that can be used to model the electron propagation in matter (ii) focus on the 3 most simple and efficient ones (iii) introduce the WP-, well known in optics but was never considered in EM, it could be an efficient way to introduce back-propagation (that was no)

- the aim of this paper was to introduce and use efficient models that could be useful in the on-going present development, especially ptychography and phase reconstruction. Much more work will be needed to see the impact the new models in phase reconstruction. (7h41)

The results point to the Fresnel free-space propagator as the main limitation of F-MS in the high-angle and thick-specimen regimes tested here. In the Au [100] ODE benchmark, the discrepancy appears first in the 135 mrad beam and only later in the central beam, after the incorrectly propagated diffracted components have had enough thickness to interfere back into it. In the Si [111] CBED calculation, the F-MS peak displacements follow the parabolic-versus-exact dispersion error, while AS-MS remains close to WP-MS over the measured angular range. In the thick Au [100] CBED calcu-

lation, the F-MS amplitude error becomes roughly an order of magnitude larger than the AS-MS error beyond about 10 nm. These observations indicate that the principal F-MS error is the accumulated longitudinal phase error from the parabolic Fresnel propagator.

AS-MS removes most of the error in the F-MS method by changing only the propagator in the propagation step. The Fresnel kernel is replaced by the exact one-way angular-spectrum kernel, while the transmission step and FFT structure are unchanged. WP-MS makes a further correction by allowing the propagation kernel to depend on the local electrostatic refractive index, so the wavelength can change with the projected potential inside each slice. This makes WP-MS the more complete one-way split-step model, and it is closest to the ODE reference in the Au [100] benchmark. Under the high-energy conditions considered here, however, this extra local-wavelength correction is smaller than the correction obtained by removing the Fresnel approximation. For these specimens and energies, the main source of error is the paraxial free-space dispersion relation rather than the absence of a potential-dependent wavelength in the propagation step.

The silicon calculation shows that the same propagator error is not limited to the most severe high- Z case. Silicon scatters more weakly than gold, so the CBED amplitude difference is smaller than in the thick Au [100] calculation. However, the F-MS peak shifts still follow the angular dispersion error, while AS-MS remains close to WP-MS across the measured range. A conventional image simulation can hide part of this error if the transfer function or objective aperture suppresses the affected Fourier components, whereas CBED and ptychographic forward models will use the diffraction-plane intensities directly and the error between methods will be more prominent.

In particular in a ptychographic reconstruction, the forward model predicts detector intensities for each probe position. If the propagator gives the wrong high-angle amplitudes, peak positions, or accumulated phase through a thick specimen, the optimization can compensate through the recovered potential. The thick Au [100] CBED calculation shows that this modeling error is much larger for F-MS than for AS-MS after approximately 10 nm. For forward models that already use F-MS, most of the wide-angle correction can be recovered by replacing $\exp(-i\pi k_{\perp}^2 \Delta z/k_0)$ with $\exp(2\pi i \Delta z[\sqrt{k_0^2 - k_{\perp}^2} - k_0])$, while keeping the same FFT structure, transmission step, and per-slice computational complexity. The sampling estimate above shows that the corresponding sampling penalty is only a few percent under the TEM conditions considered here.

The comparison here is limited to high-energy, forward-scattering calculations where the free-space dispersion error is larger than the local wavelength correction. WP-MS should be tested separately at lower accelerating voltages, for thicker high- Z specimens, and for slice thicknesses where the potential-dependent wavelength change becomes a comparable source of phase error. Its main advantage in those tests is that it provides a differentiable one-way method with a propagation kernel that depends on the local electrostatic refractive index, that can make use of the computational efficiency of the Fourier transform

5. Conclusion

We adapted the optimised Wave Propagation Method developed in light optics to electron microscopy, giving a wave propagation multislice (WP-MS) method that tracks high-angle propagation and local wavelength changes within a differentiable split-step framework, that can make efficient use of the Fourier transform. We then compared F-MS, AS-MS, and WP-MS in three complementary settings: an Au [100] ODE benchmark, a lower- Z Si [111] CBED calculation, and a thick Au [100] CBED calculation.

Across these tests, replacing the Fresnel kernel with the exact angular-spectrum kernel accounts for most of the improvement over conventional multislice. WP-MS is consistently the most complete one-way model and is closest to the ODE reference in the Au [100] benchmark, but its additional accuracy over AS-MS is small under the high-energy conditions tested here. The main practical recommendation is that for high-angle elastic forward simulations where F-MS is still being used, AS-MS should be the default standard, because it removes the dominant paraxial dispersion error without changing the basic FFT-based multislice workflow.

Future work could extend the WP-MS toward a bidirectional formulation that can include backscattering, and then benchmark it against FC-MS in lower-energy simulations, and measure how F-MS, AS-MS, and WP-MS affect ptychographic reconstructions across specimen thickness, accelerating voltage, and atomic number.

References

- [1] H. Bethe, [Theorie der Beugung von Elektronen an Kristallen](#) 392 (17) 55–129. [doi:10.1002/andp.19283921704](#).

URL <https://onlinelibrary.wiley.com/doi/10.1002/andp.19283921704>

- [2] P. B. Hirsch, A. Howie, R. B. Nicholson, D. W. Pashley, M. J. Whelan, *Electron Microscopy of Thin Crystals*, Butterworths.
- [3] J. M. Cowley, A. F. Moodie, The scattering of electrons by atoms and crystals. I. A new theoretical approach, *Acta Crystallographica* 10 (10) (1957) 609–619. doi:10.1107/S0365110X57002194.
- [4] K. Ishizuka, N. Uyeda, A new theoretical and practical approach to the multislice method, *Acta Crystallographica Section A* 33 (5) (1977) 740–749. doi:10.1107/S0567739477001879.
- [5] A. Rother, K. Scheerschmidt, Relativistic effects in elastic scattering of electrons in TEM, *Ultramicroscopy* 109 (2) (2009) 154–160. doi:10.1016/j.ultramic.2008.08.008.
- [6] S. C. Hillier, E. T. Robertson, G. D. Reid, R. D. Haynes, M. D. Robertson, On the role of the second-order derivative term in the calculation of convergent beam diffraction patterns, *Ultramicroscopy* 179 (2017) 73–80. doi:10.1016/j.ultramic.2017.04.001.
URL <https://www.sciencedirect.com/science/article/pii/S0304399117300384>
- [7] C. Wacker, R. R. Schröder, Multislice algorithms revisited: Solving the Schrödinger equation numerically for imaging with electrons, *Ultramicroscopy* 151 (2015) 211–223. doi:10.1016/j.ultramic.2014.12.008.
- [8] W. Q. Ming, J. H. Chen, Validities of three multislice algorithms for quantitative low-energy transmission electron microscopy, *Ultramicroscopy* 134 (2013) 135–143. doi:10.1016/j.ultramic.2013.04.013.
- [9] J. H. Chen, D. Van Dyck, Accurate multislice theory for elastic electron scattering in transmission electron microscopy, *Ultramicroscopy* 70 (1997) 29–44. doi:10.1016/S0304-3991(97)00071-5.
- [10] G. R. Hadley, Wide-angle beam propagation using Padé approximant operators, *Optics Letters* 17 (20) (1992) 1426–1428. doi:10.1364/OL.17.001426.

- [11] K.-H. Brenner, W. Singer, Light propagation through graded-index media using the wave propagation method, *Applied Optics* 32 (26) (1993) 4984–4988. [doi:10.1364/AO.32.004984](https://doi.org/10.1364/AO.32.004984).
- [12] S. Schmidt, T. Tiess, S. Schröter, R. Hambach, M. Jäger, H. Bartelt, A. Tünnermann, H. Gross, Wave-optical modeling beyond the thin-element approximation, *Optics Express* 24 (26) (2016) 30188–30200. [doi:10.1364/OE.24.030188](https://doi.org/10.1364/OE.24.030188).
- [13] N. Barré, M. Brunel, Differentiable wave propagation method for shape optimization of freeform optics beyond the paraxial approximation, *Optics Letters* 50 (9) (2025) 2860–2863. [doi:10.1364/OL.559067](https://doi.org/10.1364/OL.559067).
- [14] G. S. Chen, Y. T. He, W. Q. Ming, C. L. Wu, D. Van Dyck, J. H. Chen, Fast STEM image simulation in low-energy transmission electron microscopy by the accurate Chen-van-Dyck multislice method, *Micron* 190 (2025) 103778. [doi:10.1016/j.micron.2024.103778](https://doi.org/10.1016/j.micron.2024.103778).
- [15] E. J. Kirkland, *Advanced Computing in Electron Microscopy*, 2nd Edition, Springer, New York, 2010. [doi:10.1007/978-1-4419-6533-2](https://doi.org/10.1007/978-1-4419-6533-2).
- [16] J. W. Goodman, *Introduction to Fourier Optics*, 4th Edition, W. H. Freeman, New York, 2017.
- [17] K. Fujiwara, [Relativistic dynamical theory of electron diffraction](https://doi.org/10.1143/JPSJ.16.2226), *Journal of the Physical Society of Japan* 16 (11) (1961) 2226–2238. [doi:10.1143/JPSJ.16.2226](https://doi.org/10.1143/JPSJ.16.2226)
URL <https://journals.jps.jp/doi/10.1143/JPSJ.16.2226>
- [18] A.-J. Fresnel, [Mémoire sur la diffraction de la lumière](https://doi.org/10.1143/JPSJ.16.2226), prize memoir of 1818; reprinted in *Oeuvres complètes d’Augustin Fresnel*, vol. 1, Impr. Impériale, Paris, 1866, pp. 247–382 (1818).
URL https://books.google.com/books?id=110_AAAAcAAJ
- [19] J. A. Fleck, J. R. Morris, M. D. Feit, Time-dependent propagation of high energy laser beams through the atmosphere, *Applied Physics* 10 (2) (1976) 129–160. [doi:10.1007/BF00896333](https://doi.org/10.1007/BF00896333).
- [20] H. G. Booker, P. C. Clemmow, The concept of an angular spectrum of plane waves, and its relation to that of polar diagram and aperture

- distribution, Proceedings of the IEE, Part III 97 (1950) 11–17. doi:
[10.1049/pi-3.1950.0002](https://doi.org/10.1049/pi-3.1950.0002).
- [21] A. Korpel, K. E. Lonngren, P. P. Banerjee, H. K. Sim, M. R. Chatterjee, Split-step-type angular plane-wave spectrum method for the study of self-refractive effects in nonlinear wave propagation, Journal of the Optical Society of America B 3 (6) (1986) 885–890. doi:
[10.1364/JOSAB.3.000885](https://doi.org/10.1364/JOSAB.3.000885).
 - [22] P. C. Lee, E. Voges, Three-dimensional semi-vectorial wide-angle beam propagation method, Journal of Lightwave Technology 12 (2) (1994) 215–225. doi:[10.1109/50.350601](https://doi.org/10.1109/50.350601).
 - [23] D. van Dyck, Improved methods for the high speed calculation of electron microscopic structure images, Physica Status Solidi A 52 (1) (1979) 283–292. doi:[10.1002/pssa.2210520131](https://doi.org/10.1002/pssa.2210520131).
 - [24] S. Banerjee, A. Sharma, Propagation characteristics of optical waveguiding structures by direct solution of the helmholtz equation for total fields, Journal of the Optical Society of America A 6 (12) (1989) 1884–1894. doi:[10.1364/JOSAA.6.001884](https://doi.org/10.1364/JOSAA.6.001884).
 - [25] A. Sharma, S. Banerjee, Method for propagation of total fields or beams through optical waveguides, Optics Letters 14 (1) (1989) 96–98. doi:
[10.1364/OL.14.000096](https://doi.org/10.1364/OL.14.000096).